MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Three tiers of questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 7-9 (builds on Primary 2 maths)

Math Inquirers P2 Ages 7-9. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three tiers: Start here, Think harder and Go further. They are open to everyone. Nobody is placed in a tier: every child can start with Start here, and any child may try Think harder or Go further at any time. Go further questions are good ones to talk about together.
- START HERE questions are approachable for a child who has not yet been taught the topic. They can start with objects, a drawing, counting, guess and check or skills from earlier years, and they use no unexplained words. Use them for warm-ups and to build confidence.
- THINK HARDER questions need the idea taught in the lesson, used in a new setting: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- GO FURTHER questions need the topic well understood, plus an extension: find every way, work backwards, test a case, justify an answer, change the problem or make up your own. This program is inspired by research-based approaches used in gifted education; it is not an official GEP standard and is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Digit Detectives (Numbers to 1000)	6 + 8 + 9 questions
Lesson 2	Missing Digit Mysteries (Addition and Subtraction within 1000)	6 + 9 + 9 questions
Lesson 3	Legs, Wheels and Arrays (Multiplication and Division)	6 + 9 + 9 questions
Lesson 4	Pattern Machines (Patterns with Multiplication)	6 + 9 + 9 questions
Lesson 5	Mystery Number Machines (Working Backwards)	6 + 9 + 9 questions
Lesson 6	Market Day (Money)	6 + 9 + 9 questions
Lesson 7	Fair Shares (Fractions)	6 + 8 + 9 questions
Lesson 8	Shape Builders (2D and 3D Shapes)	6 + 8 + 9 questions
Lesson 9	Measure Masters (Length, Mass, Volume and Time)	6 + 10 + 9 questions
Lesson 10	Challenge Day (Mixed Puzzles)	6 + 8 + 9 questions

Lesson 1: Digit Detectives

Numbers to 1000 • Week 1

START HERE (open with objects, drawing or counting)

S1. What number is 5 hundreds, 3 tens and 8 ones?

S2. In 914, what is the digit 1 worth?

S3. What number is 100 more than 457?

S4. What number is 10 less than 603?

S5. Use the digit cards 3, 6 and 1 once each. Make the biggest 3-digit number.

S6. Put these numbers in order, smallest first: 409, 490, 94, 940.

THINK HARDER (the lesson idea in a new setting)

T1. Use the digit cards 2, 6 and 9 once each. Make a list in order of every 3-digit number you can make that is bigger than 500. How do you know you have found them all?

T2. Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?

T3. Ali has 46 ten-blocks. Siti has 3 hundred-blocks and 15 ten-blocks. Who has the bigger number? How much bigger?

T4. Mei says: “If I swap the tens digit and the ones digit of a 3-digit number, the new number is always smaller.” Is she right? Show what happens with some examples. When does the number stay the same?

T5. I am a 3-digit number. My digits add up to 4. Find all the numbers I could be.

T6. Use the digit cards 2, 5 and 9 once each. Which number you can make is closest to 600?

T7. How many 3-digit numbers have 7 in the hundreds place and 0 in the ones place?

T8. On a 1000 chart with rows of 10, which number is directly below 495 (one square down)? Which number is directly above 501 (one square up)?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Use any three of the digit cards 1, 3, 5 and 7. How many different 3-digit numbers can you make?

G2. How many 3-digit numbers have digits that add up to 5?

G3. I am an odd 3-digit number. My three digits are all different and add up to 6. Find all the numbers I could be.

G4. Make up your own 3-digit number riddle with three clues, so that exactly ONE number fits (use clues about the hundreds digit, the digits adding up, odd or even, double or half). Write the answer on the back. After each clue, say how many numbers still fit. Swap with a partner and solve theirs.

G5. A book has pages numbered 1 to 120. How many digits are printed altogether?

G6. Use three DIFFERENT cards from 0, 2, 5 and 8 to make a 3-digit number. How many different numbers can you make? Why is it not $4 \times 3 \times 2 = 24$?

G7. Mei uses each of the digit cards 1, 2, 3, 4, 5 and 6 once to make two 3-digit numbers. She adds them. What is the smallest total she can make?

G8. Ali counts in 5s, starting at 3: 3, 8, 13, 18 … Will he say 500? Will he say 998? Explain.

G9. Ben writes all the numbers from 300 to 399. How many times does he write the digit 3?

Lesson 2: Missing Digit Mysteries

Addition and Subtraction within 1000 • Week 1

START HERE (open with objects, drawing or counting)

S1. 3[ ]5 + 214 = 579. Which digit goes in the box?

S2. 6[ ]8 – 205 = 433. Which digit goes in the box?

S3. What number goes in the box to make this true? 400 + 250 = 300 + [ ]

S4. What must be added to 365 to make 500?

S5. What number goes in the box? [ ] – 150 = 600

S6. True or false? 299 + 101 = 300 + 100. Explain.

THINK HARDER (the lesson idea in a new setting)

T1. A magic triangle has 3 corner circles and 3 circles in the middle of the sides. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so that every side adds up to 420. Draw the triangle. Which numbers are in the corners?

T2. Find the missing digits: 5[ ] + [ ]7 + 38 = 132. There are three numbers, so the carry from the ones may be more than 1.

T3. Mei did 523 – 267 and wrote 266. She checked by adding 266 + 267 = 533. Was her answer right? What mistake did she make? What is the right answer?

T4. True or false? Say how you know without working out both sides. (a) 346 + 199 = 345 + 200 (b) 270 + 130 = 260 + 150 (c) 618 + 99 = 600 + 117

T5. A shop had 905 eggs. It sold some in the morning and 260 in the afternoon. 345 eggs were left. How many eggs were sold in the morning?

T6. What number goes in the box to make this true? 900 – [ ] = 450 + 175

T7. Which is bigger, 347 + 258 or 350 + 255? Explain how you know without adding.

T8. The numbers 20, 40, 60, 80, 100, 120, 140, 160 and 180 make a magic square. What is the magic total? Which number goes in the middle?

T9. Put 11, 12, 13, 14, 15 and 16 in a magic triangle so each side adds up to 39. Which numbers go in the corners?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as BIG as possible.

G2. The same number goes in both boxes: [ ] + 150 = 550 – [ ]. What is the number?

G3. I add 199 to my number, then take away 99. I get 650. What is my number?

G4. Put 110, 120, 130, 140, 150 and 160 in a magic triangle. What is the smallest side total you can make? The biggest? Find a triangle with side total 400. Could the side total be 380? Explain.

G5. Make up your own missing-digit addition with TWO missing digits (use two boxes) that has exactly ONE answer. Then make one with two missing digits that has MORE than one answer. What is different about the second one? Swap with a partner and check each other's.

G6. What is the smallest 3-digit number you can add to 386 so that the answer has three digits that are all the same?

G7. Find ALL the ways to fill in the four boxes: [ ][ ]7 + 4[ ][ ] = 1000

G8. Box A and Box B hold 250 marbles together. Box B and Box C hold 350. Box A and Box C hold 400. How many marbles are in each box?

G9. Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as SMALL as possible.

Lesson 3: Legs, Wheels and Arrays

Multiplication and Division • Week 2

START HERE (open with objects, drawing or counting)

S1. How many wheels are there on 5 bicycles?

S2. There are 4 rows of chairs with 5 chairs in each row. How many chairs are there?

S3. 18 sweets are shared equally among 3 children. How many sweets does each child get?

S4. How many legs do 7 hens have?

S5. An array has 2 rows of 10 dots. Write two multiplication sentences and two division sentences for it.

S6. 25 pencils are tied in bundles of 5. How many bundles are there?

THINK HARDER (the lesson idea in a new setting)

T1. A shop has some bicycles (2 wheels) and some tricycles (3 wheels), at least one of each. There are 17 wheels altogether. Find ALL the ways. Start with 1 tricycle, then 2 tricycles, and so on. Make a table or use counters. How do you know you have found them all?

T2. A farm has 9 animals, each a hen (2 legs) or a goat (4 legs). Mei counts 25 legs. Siti counts 26 legs. Only one of them can be right. Who? How many hens and goats?

T3. Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?

T4. An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?

T5. 27 children go on a boat ride. Each boat holds 4 children. How many boats are needed? How many empty seats are there?

T6. In a car park there are cars (4 wheels) and motorcycles (2 wheels). There are 8 vehicles and 22 wheels. How many cars are there?

T7. Find all the numbers from 21 to 49 that can be shared equally into 5 groups AND equally into 2 groups with nothing left over.

T8. Siti has 24 cupcakes. She puts them in boxes with the same number in every box. The boxes hold 2, 3, 4, 5 or 10 cupcakes. Which box sizes leave no cupcakes over, and how many boxes does each need?

T9. A table has 4 legs and a stool has 3 legs. Ravi counts 17 legs on the tables and stools in a room. How many tables and how many stools are there?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Ali says: “Every number from 4 to 30 can be made into an array with at least 2 rows and at least 2 counters in each row.” Is he right? Find ALL the numbers that cannot. Explain why they cannot.

G2. A shop sells bicycles (2 wheels), tricycles (3 wheels) and toy cars (4 wheels). There are 9 vehicles and 28 wheels altogether, with at least one of each kind. Find ALL the ways.

G3. A pen has 10 animals, all hens (2 legs) and goats (4 legs). What are ALL the numbers of legs that Mei could count? Why can she never count 21 legs, or 42 legs?

G4. Mei says: "I counted the legs of some hens and goats. There were 9 legs." Can she be right? Explain.

G5. Make up your own “how many of each” puzzle with two kinds of things that have different numbers of legs or wheels. Write the clues (how many things, how many legs or wheels) so that there is exactly ONE answer. Solve it with a table, then swap with a partner. Then change the number of legs in your puzzle so that it has NO answer.

G6. I am a number between 10 and 30. I can be put into groups of 3 with none left over, and into groups of 4 with none left over. When I am put into groups of 5, 4 are left over. What number am I?

G7. A shop has 12 vehicles. Each one is a bicycle (2 wheels) or a tricycle (3 wheels). Mei says: “Ravi could count 24 wheels, or 25, or 26, or any number of wheels up to 36.” Is she right? Could he count 37? Then think about 12 hens (2 legs) and goats (4 legs): why can Ravi count 25 wheels but never 25 legs?

G8. Mr Lim has between 20 and 30 sweets. When he puts them in groups of 4, 3 are left over. When he puts them in groups of 5, 2 are left over. How many sweets does he have?

G9. A pen has hens and goats. There are 2 more hens than goats. There are 22 legs altogether. How many hens and how many goats?

Lesson 4: Pattern Machines

Patterns with Multiplication • Week 2

START HERE (open with objects, drawing or counting)

S1. Count in 4s. Fill in the next three numbers: 4, 8, 12, __, __, __

S2. A pattern goes red, blue, red, blue ... What colour is the 10th bead?

S3. Each hand has 5 fingers. How many fingers are on 6 hands?

S4. A pattern goes star, circle, circle, star, circle, circle ... What is the 9th shape?

S5. What is the rule? 3, 7, 11, 15 ... Write the next two numbers.

S6. A number pattern goes 10, 20, 30 ... What is the 8th number?

THINK HARDER (the lesson idea in a new setting)

T1. Look at the shape pattern in the Start here page again (places 1 to 10). How many circles are there in the first 30 shapes? How do you know?

T2. Beads go red, red, blue, green, red, red, blue, green ... Mei says the 30th bead is blue. Is she right? Which places up to 30 have a green bead?

T3. Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?

T4. Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?

T5. Ali counts in 4s from 4. Mei counts in 10s from 10. Which numbers up to 50 do they both say?

T6. A pattern goes 1, 2, 3, 1, 2, 3 ... What is the total of the first 15 numbers?

T7. Ben's pattern is 2, 5, 8, 11 ... Will 30 be in his pattern? Explain.

T8. A pattern repeats every 4 shapes. The 20th shape is a heart. What is the 40th shape? How do you know?

T9. A pattern goes square, circle, circle, square, circle, circle ... Mei draws 20 shapes. How many squares does she draw?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Light A flashes every 3 seconds. Light B flashes every 5 seconds. They flash together at the start. In the next 30 seconds, when do they flash together?

G2. Houses are made of sticks in a row: 1 house uses 5 sticks, 2 houses use 9, 3 houses use 13. Ben has 40 sticks. What is the most houses he can make? How many sticks are left over?

G3. A frog starts at 1 and jumps 4 each time: 1, 5, 9, 13 ... Will it land on 37? Will it land on 30? Explain.

G4. A pattern repeats every 5 shapes. In each group, only the 3rd shape is a star. Which of these places have a star: 13, 23, 30, 48?

G5. Make up your own repeating pattern of 4 shapes. Write 3 "far place" questions about it for a partner (one place in the 40s), with the answers. Then change your pattern so it repeats every 5 shapes. Which of your answers change? Which stay the same?

G6. A pattern goes triangle, square, circle, circle, circle, triangle, square, circle, circle, circle ... How many circles are in the first 42 shapes?

G7. Ali counts in 3s from 3. Mei counts in 4s from 4. They both say 12. Predict the next number they will BOTH say. Then list every number up to 100 that they both say. What do you notice? Why?

G8. Siti builds towers of cubes. Tower 1 has 2 cubes, tower 2 has 5 cubes, tower 3 has 8 cubes. The towers keep growing in the same way. Which tower has 29 cubes? Is there a tower with exactly 50 cubes? With exactly 100 cubes? Explain how you know.

G9. Frog A starts at 0 and jumps 4 each time: 4, 8, 12 … Frog B starts at 2 and jumps 3 each time: 5, 8, 11 … Which numbers up to 30 do both frogs land on?

Lesson 5: Mystery Number Machines

Working Backwards • Week 3

START HERE (open with objects, drawing or counting)

S1. I think of a number. I multiply it by 5. I get 45. What is my number?

S2. I think of a number. I divide it by 2. I get 8. What is my number?

S3. I think of a number. I add 35. I get 90. What is my number?

S4. I think of a number. I multiply it by 3, then add 2. I get 20. What is my number?

S5. I think of a number. I take away 10, then divide by 4. I get 5. What is my number?

S6. Mei had some stickers. She gave away 14 and has 27 left. How many did she have at first?

THINK HARDER (the lesson idea in a new setting)

T1. A machine multiplies by a number, then adds a number. IN 2 gives OUT 11 and IN 5 gives OUT 20. Find what the machine does (try × 2, × 3, × 4 … with IN 2, then test IN 5). Then: what OUT does IN 10 give? Which IN gives OUT 26?

T2. Ben had some money. He spent $15 on a book. Then his dad gave him as much as he had left. Now he has $42. How much did Ben have at first? Draw bars.

T3. Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?

T4. Ali, Ben and Siti have 40 marbles altogether. Ben has 5 more marbles than Ali. Siti has 8 more marbles than Ali. How many marbles does each child have? Draw a bar for each child.

T5. A machine does × 2, then − 3. What goes IN to give OUT 17? What goes IN to give OUT 1?

T6. I think of a number. I divide it by 3, then multiply by 10. I get 50. What is my number?

T7. A bus had some people on it. At the first stop, 12 got off and 5 got on. At the second stop, 8 got off. Now there are 20 people on the bus. How many were on the bus at first?

T8. A machine does × 4, then + 13. The OUT is 45. What went IN?

T9. A machine multiplies by a number, then adds a number. IN 1 gives OUT 9. IN 3 gives OUT 19. What does the machine do?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Ali, Ben and Siti share 42 stickers. Ben gets 2 more than Ali. Siti gets twice as many as Ali. How many stickers does each child get?

G2. A machine does × 2, then − 4. Ravi puts a number in, then puts the OUT straight back in again. The final OUT is 12. What number did he put in first?

G3. Ben gives Ali 5 marbles. Now they both have 20 marbles. How many marbles did each boy have before? How many MORE marbles did Ben have than Ali at first? Why is it not 5? What if Ben had given 8 marbles and they ended with the same number?

G4. A machine does × 2, then + 1. Can the OUT ever be 20? Which OUT numbers between 10 and 20 can it make? Explain how you know.

G5. Make up your own before-and-after story: someone gives some away, then gets some more, and ends with 40. Draw the bars and work out how many they had at first. Check by going forwards. Then change ONE number in your story. What happens to the number at the start? Swap with a partner.

G6. I think of a number. If I add 10 to it, I get the same answer as when I multiply it by 3. What is my number?

G7. Siti had some sweets. She gave Ben 1 more than half of them. Now she has 7 sweets left. How many sweets did she have at first?

G8. A machine does × 2, then − 1. Mei puts a number through the machine 3 times in a row (each OUT goes back in). The final OUT is 17. What number did she start with?

G9. Three boxes have 30 pens altogether. 5 pens are moved from box A to box B. Then 3 pens are moved from box B to box C. Now all three boxes have the same number of pens. How many pens were in each box at first?

Lesson 6: Market Day

Money • Week 3

START HERE (open with objects, drawing or counting)

S1. Write 275¢ in dollars and cents.

S2. Pay exactly 65¢ with the fewest coins.

S3. Pay exactly $17 with the fewest notes.

S4. A pencil costs 45¢. Mei pays with a $1 coin. How much change does she get?

S5. Ali has $20. He buys a book for $12. Does he have enough money left to buy a kite for $9?

S6. Ben had some money. He spent $6 and has $9 left. How much money did he have at first?

THINK HARDER (the lesson idea in a new setting)

T1. Ben spent half of his money on a kite. Then he spent $4 on a ball. He had $6 left. How much money did Ben have at first? Tip: draw his money as a bar cut in half.

T2. How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?

T3. Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.

T4. How many different ways can you pay exactly 40¢ using 5¢, 10¢ and 20¢ coins? Make a list in order, starting with the most 20¢ coins.

T5. Pens cost $3 each and books cost $7 each. Ben buys 6 things and spends exactly $30. How many pens and how many books does he buy? Make a table.

T6. Ravi had some money. He spent $8, then he spent half of what was left. He had $6 left. How much money did Ravi have at first?

T7. Siti has 5 coins. They are all 20¢ or 50¢ coins, worth $1.60 altogether. How many of each coin does she have?

T8. Ali pays exactly 90¢ with 5 coins. Find all the sets of 5 coins he could use.

T9. A pen costs $3 more than a pencil. Together they cost $11. How much does each cost?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Ali has 10 coins. Every coin is a 10¢ coin or a 20¢ coin. They could be worth $1.50: how many of each coin is that? Write ALL the totals his 10 coins could be worth. Could they be worth $1.55? Explain.

G2. Mei has $1. At the snack stall: chips 45¢, sweet 15¢, jelly 25¢, biscuit 30¢, lollipop 20¢. She wants as many DIFFERENT snacks as she can. Which should she buy?

G3. Ben has 6 coins. Every coin is 10¢ or 50¢. List all the amounts he could have.

G4. Which amounts 10¢, 20¢, 30¢, … up to $1 can be paid exactly using only 20¢ and 50¢ coins? Which cannot? Explain why.

G5. Make up your own price list for 5 snacks (prices in cents) so that Mei, with exactly $1, can buy exactly TWO different baskets that cost exactly $1. Check by listing every basket in order. Swap with a partner. Is your answer the same?

G6. Three friends share the cost of a $15 present. Ali pays $2 more than Ben. Siti pays $2 more than Ali. How much does each pay?

G7. Siti had some money. She spent half of it on a book. Then she spent half of what was left on a pen. She had $5 left. How much money did she have at first?

G8. Sweets cost 5¢ and lollipops cost 20¢. Ben buys 8 things, at least one of each kind. Mei says he spent $1.00. Siti says he spent $1.50. Could either of them be right? Find ALL the amounts Ben could have spent. What do you notice about the amounts, and why?

G9. Mei has one $10 note, one $5 note, one $2 note and one $1 coin. How many different amounts can she pay exactly, using one or more of them?

Lesson 7: Fair Shares

Fractions • Week 4

START HERE (open with objects, drawing or counting)

S1. A cake is cut into 5 equal pieces. Ben eats 2 pieces. What fraction of the cake does he eat?

S2. How many quarters make 1 whole?

S3. Which is bigger, 1/3 or 1/6?

S4. 3/10 + 4/10 = ?

S5. Which is smaller, 5/8 or 3/8?

S6. A pizza is cut into 6 equal slices. 4 slices are eaten. What fraction of the pizza is left?

THINK HARDER (the lesson idea in a new setting)

T1. Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.

T2. Mei says: “3/8 of a pizza is more than 3/5 of the same pizza, because 8 is bigger than 5.” Is she right? Draw two bars of the same length to show.

T3. Siti says: “3/8 + 2/8 = 5/16, because 8 + 8 = 16.” Is she right? Draw a bar to show. What is the right answer, and why does the bottom number not change?

T4. Ali says: “Half of my pizza is bigger than half of your pizza.” Can he be right? Explain.

T5. Draw 3 squares. Cut each one into 4 EQUAL parts in a different way.

T6. Ravi eats 3/8 of a pizza. Ben eats 2/8 of the same pizza. How much more does Ravi eat? How much of the pizza is left?

T7. A strip is cut into 4 equal parts. Then each part is cut in half. How many parts are there now? What is each part called?

T8. Siti shades 4/12 of a shape. Then she shades 5/12 more. How much of the shape is NOT shaded?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A paper strip is folded in half, then in half again, then in half again. When you open it, how many equal parts are there? What fraction is each part?

G2. Siti eats 1/2 of a pizza. Then Ravi eats 1/2 of what is left. What fraction of the whole pizza is left now?

G3. Two fractions with bottom number 9 add up to 1 whole. One of them is 3/9 more than the other. What are they?

G4. How many different ways can you add two fractions with bottom number 6 to make 1 whole? Each fraction must be at least 1/6, and 1/6 + 5/6 and 5/6 + 1/6 count as the same way.

G5. Make up your own fair-sharing story: a cake is cut into 10 equal pieces. Three people eat different amounts and 2/10 is left. Write each person’s share as a fraction, draw the bar, and check that the shares and what is left make 1 whole. Swap with a partner.

G6. Ali says: “The bigger the bottom number, the bigger the fraction.” Is he right? Use bars of the same length to show that he is wrong. Then say how you pick the bigger fraction when two fractions have the same bottom number.

G7. A cake is cut into 12 equal pieces. Mei gets 4/12. Ali and Ben share the rest, but Ali gets 2/12 more than Ben. What fraction does Ben get?

G8. Three children share a cake cut into 10 equal pieces. Each child gets a DIFFERENT number of pieces (at least 1), and the whole cake is shared out. Find ALL the ways to share it. Write each child's share as a fraction.

G9. I am a fraction with 1 on top. I am smaller than 1/4 but bigger than 1/7. What could I be? Find all the answers.

Lesson 8: Shape Builders

2D and 3D Shapes • Week 4

START HERE (open with objects, drawing or counting)

S1. How many quarter circles make a whole circle?

S2. How many half circles make a whole circle?

S3. How many faces does a cube have?

S4. How many squares of ALL sizes are in a grid 2 squares long and 2 squares high? (Small ones and one big one.)

S5. A tower has 3 cubes. Ali builds 2 more towers just like it. How many cubes are there altogether?

S6. Which solid has 2 flat faces that are circles and 1 curved face?

THINK HARDER (the lesson idea in a new setting)

T1. A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.

T2. How many rectangles can you find in a grid 2 squares long and 2 squares high? Remember that a square is a rectangle too. Count them by size.

T3. Cut out nets E and F from the cut-out page and try folding them. Which one folds up into a cube? How do you know?

T4. Six squares are joined in a straight line. Can they fold into a cube? Explain.

T5. A shape is made of a square with a half circle on each of its 4 sides, curved sides outward. How many straight sides does its outside edge have? How many curved sides?

T6. How many rectangles can you find in a row of 3 squares joined side by side? (A square is a rectangle too. Count all sizes.)

T7. Mei builds a stack 3 cubes long, 2 wide and 3 high. She takes away the whole top layer. Then she takes away the 4 corner cubes of the new top layer. How many cubes are left?

T8. A stack is 4 cubes long, 2 cubes wide and 2 cubes high. How many cubes?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. How many squares can you find in a grid 4 squares long and 3 squares high? Another grid, longer than it is high and at least 2 squares high, also has exactly 20 squares of all sizes. Find ALL the grids like that.

G2. A square has both lines from corner to corner drawn. How many triangles can you find?

G3. Design your own cube stack: draw a stack with at least 2 layers that has exactly 20 cubes. Then change it into a different stack that also has exactly 20 cubes. Write how many cubes are in each layer. Swap with a partner and check each other's.

G4. A big cube is built from small cubes: 3 long, 3 wide and 3 high. Its outside is painted. How many small cubes have no paint at all?

G5. Steps are made of cubes. They are 3 cubes high, then 2, then 1, and every step is 3 cubes wide from front to back. How many cubes?

G6. How many triangles are in a triangle with 3 lines from the top corner to the bottom and 1 line across the middle?

G7. A pyramid of cubes has a top layer of 1 cube, then a layer 2 by 2, then a layer 3 by 3, then a bottom layer 4 by 4. How many cubes are there?

G8. How many squares can you find in a grid 5 squares long and 5 squares high?

G9. A 3 by 3 by 3 cube is built from 27 small cubes and the whole outside is painted. Sort the small cubes: how many have paint on 3 faces, on 2 faces, on 1 face and on no faces? Check that your four numbers make 27. What if the big cube is 4 by 4 by 4: how many small cubes have no paint?

Lesson 9: Measure Masters

Length, Mass, Volume and Time • Week 5

START HERE (open with objects, drawing or counting)

S1. One bag of flour balances 2 bags of sugar. A bag of sugar is 300 g. How heavy is the bag of flour?

S2. A bucket holds 10 ℓ. It has 4 ℓ of water in it. Ali pours in 3 ℓ more. How many more litres are needed to fill it?

S3. A ribbon is 90 cm long. Mei cuts off 25 cm. How long is the piece that is left?

S4. A cartoon starts at 5:40 pm and lasts 30 minutes. What time does it end?

S5. Write am or pm. Ben eats lunch at 12:30 ____. He goes to bed at 8:45 ____.

S6. The minute hand points to 7. How many minutes past the hour is it?

THINK HARDER (the lesson idea in a new setting)

T1. 2 apples and 1 lemon balance 1 pomelo. 1 pomelo balances 5 lemons. How many lemons balance 1 apple? How many lemons balance 1 pomelo and 1 apple together?

T2. Mei's family leaves home at 10:45 am. They reach Grandma's flat at 12:10 pm. How long is the trip?

T3. 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?

T4. You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.

T5. Mei's piano lesson ends at 4:15 pm. It lasted 45 minutes. What time did it start?

T6. Buses to the MRT station leave every 15 minutes. The first bus leaves at 6:50 am. Ravi reaches the bus stop at 7:40 am. How long does he wait? The ride takes 25 minutes. When does he reach the MRT station?

T7. Three ribbons, each 30 cm long, are glued end to end in a line. Each join overlaps by 2 cm. How long is the long ribbon?

T8. A tank has 20 ℓ of water. A tap adds 5 ℓ every 10 minutes. The tap is turned on at 9:40 am. At what time does the tank have 40 ℓ?

T9. A box with 4 same books in it is 1000 g. The empty box is 200 g. How heavy is one book?

T10. Ali's clock is 10 minutes fast. It shows 3:05 pm. What is the real time?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A basket with 4 apples is 700 g. The same basket with 2 apples is 500 g. How heavy is the empty basket?

G2. A rope 40 cm long is cut into pieces 10 cm long. Each cut takes 2 minutes. How long does the cutting take?

G3. A film is 1 hour 40 minutes long. It ends at 12:20 pm. What time did it start?

G4. Ali's watch loses 5 minutes every hour. He sets it to the right time at 8:00 am. What does his watch show at 12 noon?

G5. Make up your own jug puzzle: choose two jug sizes between 2 ℓ and 9 ℓ and a target amount, and check that it can be done by listing every pour. Then try this: with a 4 ℓ jug and a 6 ℓ jug, can you ever measure exactly 5 ℓ? Explain why or why not.

G6. Ravi has a 5 ℓ jug, a 3 ℓ jug and a big bucket. How can he get exactly 7 ℓ of water in the bucket?

G7. You have a 3 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 5 ℓ? List each pour.

G8. Scale 1: 2 cubes and 1 ball balance 2 balls. Scale 2: 1 ball and 1 cube balance a 300 g weight. How heavy is one cube? How heavy is one ball?

G9. Mei has 3 lessons in a row. Each lesson is 35 minutes long, with a 10-minute break between lessons. The first lesson starts at 8:15 am. What time does the last lesson end?

Lesson 10: Challenge Day

Mixed Puzzles • Week 5

START HERE (open with objects, drawing or counting)

S1. Use the digit cards 5, 1 and 9 once each. Make the greatest 3-digit number and the smallest 3-digit number.

S2. In a picture graph, each picture stands for 2 children. Apples has 4 pictures. How many children chose apples?

S3. Ali has 3 bags with 5 marbles in each bag. Mei has 4 bags with 4 marbles in each bag. Who has more marbles? How many more?

S4. Make 80¢ with exactly 3 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins.

S5. A cake is cut into 8 equal pieces. Siti eats 2 pieces and Ben eats 3 pieces. What fraction of the cake is left?

S6. Ben is taller than Ali. Ali is taller than Mei. Siti is taller than Ben. Who is the tallest? Who is the shortest?

THINK HARDER (the lesson idea in a new setting)

T1. 30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?

T2. Mei saves $3 on day 1, $6 on day 2, $9 on day 3, and so on ($3 more each day). Ali saves $10 every day. Make a table of how much each child has saved altogether after each day. On which day does Mei FIRST have more than Ali?

T3. A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.

T4. 5, 9, 13, 17, ... Will 49 be in this pattern? Will 50? How do you know?

T5. A farm has chickens and cows. There are 8 heads and 22 legs. How many cows are there?

T6. Mei spent half of her money on a book. Then she spent $4 on a pen. She has $6 left. How much money did she have at first?

T7. Ali, Ben and Siti each chose a different colour: red, blue or green. Ali did not choose red. Siti chose blue. Who chose green?

T8. A rope 30 m long is cut into pieces 5 m long. How many pieces are there? How many cuts?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Use three DIFFERENT digit cards to make 3-digit numbers. With the cards 0, 3 and 8 you can make exactly 4 numbers. With the cards 3, 5 and 8 you can make 6. Which sets of three different cards make exactly 4 numbers? Test some sets and write a rule. Does your rule always work?

G2. Ali, Mei, Siti and Ben each go to school a different way: bus, MRT, car or walking. Clues: Siti walks. Ben does not go by bus or by car. Mei does not go by car. Who goes by what? Then take away ONE clue at a time. Which clue could you take away and still have exactly one answer? How many answers does each clue-less puzzle have?

G3. A cake is cut into 12 equal slices and shared by Mei, Ali and Ben. Ben gets 2 more slices than Mei, and Ali gets 2 more slices than Ben. What fraction of the cake does each child get? Each slice weighs 100 g. How heavy is Ali's share?

G4. Ali saves 20¢ on day 1, 40¢ on day 2, 60¢ on day 3, and so on, saving 20¢ more each day. On which day will the TOTAL he has saved first be MORE than $2? Make a table of the day, the amount saved that day, and the total so far.

G5. Make up your own challenge puzzle for next year's group. It can be about digits, money, patterns, fractions, shapes or logic. Write the puzzle, its answer, and a hint that says how to start (draw it, list it or make a table). Check that it has exactly ONE answer. Swap with a partner: can they solve it using your hint?

G6. A ribbon is cut into 10 equal pieces. Siti uses 3/10 of it and Ben uses 4/10 of it. What fraction is left? Each piece is 5 cm long. How long is the part that is left?

G7. A picture graph has lost its key. Football has 6 pictures and swimming has 4 pictures. 10 more children chose football than swimming. How many children chose swimming? Football?

G8. Ali, Ben, Mei and Siti each have one coin: 10¢, 20¢, 50¢ or $1. Mei has more money than Ali but less than Ben. Ben does not have the $1 coin. Who has which coin?

G9. At a hawker centre, chicken rice costs $4 a plate and noodles cost $3 a bowl. Ravi's family buys 7 dishes for $25. How many plates of chicken rice do they buy? What is the smallest total 7 dishes can cost? The biggest? Can 7 dishes cost $30? Can they cost EVERY whole-dollar amount in between?

Answer key (tutor only)

Lesson 1: Digit Detectives

S1. 538

S2. 10 (1 ten)

S3. 557

S4. 593

S5. 631

S6. 94, 409, 490, 940

T1. 4 numbers: 629, 692, 926, 962. The hundreds digit must be 6 or 9 (a number that starts with 2 is smaller than 500). With 6 in the hundreds, the other two cards make 629 and 692; with 9 in the hundreds, they make 926 and 962.

T2. 3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.

T3. Ali: 46 tens is 460. Siti: 3 hundreds and 15 tens is 300 + 150 = 450. Ali has the bigger number, by 10.

T4. No. 452 becomes 425, which is smaller, but 425 becomes 452, which is bigger. It gets smaller when the tens digit is bigger than the ones digit, bigger when it is smaller, and stays the same when the two digits are the same (733).

T5. 10 numbers: 103, 112, 121, 130, 202, 211, 220, 301, 310, 400.

T6. 592 (only 8 away).

T7. 10: 700, 710, 720, … 790.

T8. 505 is directly below 495. 491 is directly above 501.

G1. 24. Leave out one card (4 choices); each set of 3 cards makes 6 numbers: 6 + 6 + 6 + 6 = 24.

G2. 15. Hundreds digit 1: 5 numbers; 2: 4; 3: 3; 4: 2; 5: 1. 5 + 4 + 3 + 2 + 1 = 15.

G3. 6 numbers: 105, 123, 213, 231, 321, 501.

G4. Open. Example: “I am between 300 and 400. My digits add up to 9. My ones digit is double my tens digit.” Clue 1 leaves 100 numbers, clue 2 leaves 7 (306, 315, 324, 333, 342, 351, 360), clue 3 leaves only 324. Check that a partner cannot find a second number that fits all the clues.

G5. 252. Pages 1 to 9: 9 digits. Pages 10 to 99: 90 pages × 2 = 180 digits. Pages 100 to 120: 21 pages × 3 = 63 digits. 9 + 180 + 63 = 252.

G6. 18. Only 2, 5 or 8 can go in the hundreds place (3 choices), then 3 cards are left for the tens (3 choices) and 2 for the ones (2 choices): 3 × 3 × 2 = 18. 24 counts numbers that start with 0 (like 025), and those are not 3-digit numbers.

G7. 381, for example 135 + 246. The smallest digits 1 and 2 go in the hundreds places, 3 and 4 in the tens, 5 and 6 in the ones: 300 + 70 + 11 = 381.

G8. He says only numbers ending in 3 or 8. So he will not say 500, but he will say 998.

G9. 120. Hundreds place: 100 times. Tens place (330 to 339): 10 times. Ones place (303, 313, … 393): 10 times. 100 + 10 + 10 = 120.

Lesson 2: Missing Digit Mysteries

S1. 6 (365 + 214 = 579)

S2. 3 (638 − 205 = 433)

S3. 350

S4. 135

S5. 750

S6. True. 299 is 1 less than 300 and 101 is 1 more than 100, so both sides are 400.

T1. Corners 140, 150, 160. Between 140 and 150: 130. Between 150 and 160: 110. Between 160 and 140: 120 (each side: 140 + 130 + 150 = 420, 150 + 110 + 160 = 420, 160 + 120 + 140 = 420).

T2. 57 + 37 + 38 = 132. Ones: 7 + 7 + 8 = 22, so write 2 and carry 2 tens. Tens: 5 + 3 + 3 + 2 = 13. The first box is 7 and the second box is 3.

T3. No. Her check gives 533, not 523, so 266 is wrong. The right answer is 256 (ones: 13 − 7 = 6; tens: the 2 tens lent 1 ten to the ones, so 1 ten is left and we borrow again: 11 − 6 = 5; hundreds: 4 − 2 = 2). Check: 256 + 267 = 523. Mei’s tens digit is 1 too big: she forgot that the tens had already lent 1 ten to the ones.

T4. (a) True: 346 is 1 more than 345 and 199 is 1 less than 200. (b) False: 270 is 10 more than 260, but 130 is 20 less than 150, so the sides are 400 and 410. (c) True: 618 is 18 more than 600 and 99 is 18 less than 117 (both sides are 717).

T5. 300. Work backwards: 345 + 260 = 605 eggs after the morning sales; 905 − 605 = 300.

T6. 275 (450 + 175 = 625 and 900 − 625 = 275).

T7. They are the same (both 605). 347 is 3 less than 350 and 258 is 3 more than 255, so moving 3 from one number to the other does not change the total.

T8. 300 (all nine add up to 900, shared equally by the 3 rows). The middle is 100.

T9. 11, 12 and 13. Between 11 and 12: 16. Between 12 and 13: 14. Between 13 and 11: 15.

G1. 531: 654 − 123.

G2. 200 (200 + 150 = 350 and 550 − 200 = 350).

G3. 550. Work backwards: 650 + 99 = 749, then 749 − 199 = 550.

G4. Smallest 390 (corners 110, 120, 130) and biggest 420 (corners 140, 150, 160). For 400 use corners 110, 130 and 150, with 160 between 110 and 130, 120 between 130 and 150, and 140 between 150 and 110. 380 is impossible: the three corners are counted twice, so the 3 sides add up to 810 plus the corners, and the corners add up to at least 360, so each side is at least 390.

G5. Open. Example with one answer: 4[ ]6 + 37[ ] = 801 (426 + 375; ones 6 + 5 = 11 forces the second box, then the tens force the first). Example with more than one answer: [ ][ ]7 + 4[ ][ ] = 1000, where the tens digits only need to add up to 9, so there are 10 answers. When two missing digits in the same column only need to add up to a fixed number, many pairs of digits work.

G6. 169 (386 + 169 = 555). 444 is only 58 more than 386, and 58 is not a 3-digit number.

G7. 10 ways: 507 + 493, 517 + 483, 527 + 473, 537 + 463, 547 + 453, 557 + 443, 567 + 433, 577 + 423, 587 + 413, 597 + 403. Ones: 7 + 3 = 10. Hundreds: 5 + 4 + 1 = 10. The two tens digits add up to 9 (with the carried 1 they make 10).

G8. Box A 150, Box B 100, Box C 250. Box B + Box C is 100 more than Box A + Box B, so Box C has 100 more than Box A. Box A and Box C together are 400: take away the extra 100 to get 300, and half of 300 is 150. So Box A has 150 and Box C has 250. Then Box B has 250 − 150 = 100.

G9. 47: 412 − 365. The hundreds digits are next to each other (4 and 3); then make the number starting with 4 as small as possible (412) and the one starting with 3 as big as possible (365).

Lesson 3: Legs, Wheels and Arrays

S1. 10 wheels (2, 4, 6, 8, 10).

S2. 20 chairs (4 × 5 = 20).

S3. 6 sweets (18 ÷ 3 = 6).

S4. 14 legs (7 × 2 = 14).

S5. 2 × 10 = 20, 10 × 2 = 20, 20 ÷ 2 = 10, 20 ÷ 10 = 2.

S6. 5 bundles (25 ÷ 5 = 5).

T1. 3 ways: 7 bicycles and 1 tricycle (14 + 3); 4 bicycles and 3 tricycles (8 + 9); 1 bicycle and 5 tricycles (2 + 15). 0, 2 or 4 tricycles leave an odd number of wheels, which bicycles cannot make, and 6 or more tricycles need 18 wheels or more.

T2. Siti is right: 5 hens and 4 goats (10 + 16 = 26). Mei cannot be right: every hen has 2 legs and every goat has 4 legs, so the total is always an even number, and 25 is odd. (Start with 9 hens, 18 legs; each swap to a goat adds 2: 18, 20, 22, 24, 26.)

T3. 8 each (4 × 8 = 32) and 2 left over.

T4. 16 stickers. 3 rows have 12, so each row has 4 (12 ÷ 3 = 4); 4 rows of 4 is 16.

T5. 7 boats (6 boats hold 24, and the other 3 children need a boat too). 7 boats have 28 seats, so 1 empty seat.

T6. 3 cars (and 5 motorcycles). Start with 8 motorcycles (16 wheels); each swap to a car adds 2 wheels; 3 swaps make 22. Check: 12 + 10 = 22.

T7. 30 and 40 (they are in both the 5 and the 2 times tables).

T8. Boxes of 2 (12 boxes), boxes of 3 (8 boxes), boxes of 4 (6 boxes). Boxes of 5 and 10 leave 4 over.

T9. 2 tables and 3 stools (8 + 9 = 17). No other mix works: 0, 1, 3 or 4 tables leave 17, 13, 5 or 1 legs, which are not in the 3 times table.

G1. No. 5, 7, 11, 13, 17, 19, 23 and 29 cannot (8 numbers): only 1 row works.

G2. 3 ways (bikes, trikes, cars): 3, 2, 4; 2, 4, 3; 1, 6, 2.

G3. 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40 (11 numbers). All hens is 20 legs and each swap to a goat adds 2, up to all goats at 40. 21 is odd and the total is always even; 42 is more than 10 goats can have (40).

G4. No. Hens have 2 legs and goats have 4, so the total is always an even number (in the 2 times table). 9 is odd.

G5. Open. Example: spiders (8 legs) and birds (2 legs): 5 animals and 28 legs gives 3 spiders and 2 birds (24 + 4 = 28), and it is the only answer because each swap changes the legs by 6. With 5 animals and 25 legs there is no answer: the total is always even.

G6. 24. Numbers in both the 3 and 4 times tables between 10 and 30 are 12 and 24; 12 in groups of 5 leaves 2, 24 leaves 4.

G7. Mei is right. All 12 bicycles make 24 wheels and all 12 tricycles make 36. Each swap of a bicycle for a tricycle adds exactly 1 wheel, so the totals go up 1 at a time and none is skipped: every number from 24 to 36, and 37 is too many. For hens and goats each swap adds 2 legs, so the total can only be 24, 26, 28 ... 48: always even, so never 25.

G8. 27. Groups of 4 with 3 left: 23 or 27. Groups of 5 with 2 left: 22 or 27. Only 27 is in both lists.

G9. 5 hens and 3 goats. Guess and check: 1 goat and 3 hens make 10 legs; 2 and 4 make 16; 3 and 5 make 22. Check: 12 + 10 = 22.

Lesson 4: Pattern Machines

S1. 16, 20, 24

S2. Blue (every even place is blue).

S3. 30 fingers (5, 10, 15, 20, 25, 30).

S4. A circle. 9 ends a group of 3, and each group ends with a circle.

S5. Add 4. The next numbers are 19 and 23.

S6. 80

T1. 12 circles (30 shapes is 6 groups of 5, with 2 circles in each group).

T2. Mei is not right. The pattern repeats every 4 beads and 28 ends a group (4 × 7 = 28), so the 29th bead is red and the 30th bead is red. Green beads are the 4th bead of each group: places 4, 8, 12, 16, 20, 24 and 28.

T3. 12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).

T4. 25 sticks. The rule is add 4: 5, 9, 13, 17, 21, 25.

T5. 20 and 40.

T6. 30. 15 numbers is 5 groups, and each group adds up to 6: 6 + 6 + 6 + 6 + 6 = 30.

T7. No. The pattern goes 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32 ... It jumps from 29 to 32. (Every number is 1 less than a number in the 3 times table, and 30 is in the 3 times table.)

T8. A heart. 20 and 40 are both in the 4 times table, so both are the last shape of a group.

T9. 7 squares (in places 1, 4, 7, 10, 13, 16, 19). 18 shapes is 6 groups with 6 squares, and place 19 is one more square.

G1. At 15 seconds and at 30 seconds (numbers in both the 3 and 5 times tables).

G2. 9 houses use 37 sticks, with 3 left over (10 houses would need 41).

G3. It lands on 37 but not on 30. The frog lands on numbers 1 more than the 4 times table: 36 + 1 = 37, but 30 is not 1 more than a 4 times table number (it lands on 29 and 33).

G4. 13, 23 and 48. Stars are 2 more than the 5 times table: 13 = 10 + 3, 23 = 20 + 3, 48 = 45 + 3. 30 ends a group, so it is the 5th shape.

G5. Open. Example: square, circle, circle, triangle. Place 41 is a square (40 ends a group), place 46 is a circle, place 48 is a triangle. With a group of 5 the answers change, except for places that are in both the 4 and the 5 times tables (20 and 40 are still the end of a group, but the last shape of the new pattern may be different).

G6. 24 circles. 40 shapes is 8 groups with 3 circles each (24); places 41 and 42 are a triangle and a square.

G7. 24 is next. They both say 12, 24, 36, 48, 60, 72, 84 and 96 (8 numbers): the numbers go up by 12 each time, because 12 is the first number in both the 3 and 4 times tables, and each next one is 12 more.

G8. Tower 10 has 29 cubes (the rule is add 3: 2, 5, 8, 11, ... 29). Tower 17 has 50 cubes. No tower has 100: every tower has 1 less than a number in the 3 times table (tower 10: 30 − 1 = 29), and 101 is not in the 3 times table (99 and 102 are).

G9. 8 and 20. Frog A: 4, 8, 12, 16, 20, 24, 28. Frog B: 5, 8, 11, 14, 17, 20, 23, 26, 29.

Lesson 5: Mystery Number Machines

S1. 9

S2. 16

S3. 55

S4. 6 (20 − 2 = 18, 18 ÷ 3 = 6)

S5. 30 (5 × 4 = 20, 20 + 10 = 30)

S6. 41 (27 + 14)

T1. × 3, then + 5. IN 10 gives OUT 35. OUT 26 comes from IN 7 (26 − 5 = 21, 21 ÷ 3 = 7).

T2. $36. Undo the last step: dad gave him as much as he had left, so $42 is two equal bars of $21. So Ben had $21 after the book, and $21 + $15 = $36 at first. Check: 36 − 15 = 21, 21 + 21 = 42.

T3. 36 (30 − 12 = 18 is half, 18 + 18 = 36)

T4. Ali 9, Ben 14, Siti 17 (40 − 5 − 8 = 27, 27 ÷ 3 = 9; check 9 + 14 + 17 = 40)

T5. 10, and 2 (17 + 3 = 20, 20 ÷ 2 = 10; 1 + 3 = 4, 4 ÷ 2 = 2)

T6. 15 (50 ÷ 10 = 5, 5 × 3 = 15)

T7. 35 (20 + 8 = 28, 28 − 5 = 23, 23 + 12 = 35)

T8. 8 (45 − 13 = 32, 32 ÷ 4 = 8)

T9. × 5, then + 4 (1 × 5 + 4 = 9 and 3 × 5 + 4 = 19; × 2, × 3, × 4 and × 10 do not fit both)

G1. Ali 10, Ben 12, Siti 20 (40 ÷ 4 = 10)

G2. 6 (back twice: 12 → 8 → 6)

G3. Ben 25, Ali 15. Ben had 10 more, because Ben lost 5 AND Ali gained 5. If Ben gave 8, he had 16 more at first: the gap at the start is always double the number given.

G4. No. × 2 always gives an even number, and + 1 makes it odd, so every OUT is odd. The OUT numbers between 10 and 20 are 11, 13, 15, 17 and 19 (IN 5, 6, 7, 8, 9).

G5. Open. Example: Ben had some stamps, gave 18 away, then bought 13 more and has 40. Undo the last step: 40 − 13 = 27; undo the first: 27 + 18 = 45. Check: 45 − 18 = 27, 27 + 13 = 40. If the 13 changes to 15, the start changes to 43: a bigger amount bought means a smaller start.

G6. 5 (5 + 10 = 15 and 5 × 3 = 15)

G7. 16 (7 + 1 = 8 is half, 8 + 8 = 16; she gave Ben 9)

G8. 3 (17 + 1 = 18, 18 ÷ 2 = 9; 9 + 1 = 10, 10 ÷ 2 = 5; 5 + 1 = 6, 6 ÷ 2 = 3; check 3 → 5 → 9 → 17)

G9. A 15, B 8, C 7 (each ends with 10; undo the moves from the end)

Lesson 6: Market Day

S1. $2.75

S2. 50¢ + 10¢ + 5¢ (3 coins)

S3. $10 + $5 + $2 (3 notes)

S4. 55¢

S5. No. He has $8 left, which is $1 short.

S6. $15

T1. $20 ($6 + $4 = $10 is the half he kept after the kite, so the other half is also $10: $10 + $10 = $20).

T2. 10 ways (1 + 3 + 6)

T3. 2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢

T4. 9 ways: two 20¢ (1 way); one 20¢ and 20¢ more (3 ways); no 20¢ (5 ways).

T5. 3 pens and 3 books ($9 + $21 = $30). Table: 6 pens $18, 5 pens and 1 book $22, 4 and 2 $26, 3 and 3 $30 ✓. Each swap of a pen for a book adds $4.

T6. $20 ($6 + $6 = $12 was left after the $8; $12 + $8 = $20)

T7. Three 20¢ and two 50¢ (60¢ + 100¢ = 160¢)

T8. 3 sets: 50¢ + 20¢ + 10¢ + 5¢ + 5¢; 50¢ + 10¢ + 10¢ + 10¢ + 10¢; 20¢ + 20¢ + 20¢ + 20¢ + 10¢

T9. Pencil $4, pen $7 ($11 − $3 = $8, $8 ÷ 2 = $4)

G1. Five and five. Totals $1.00 to $2.00, 10¢ apart (11). Not $1.55.

G2. Sweet, lollipop, jelly and biscuit (90¢). Five snacks cost $1.35.

G3. 7 amounts: 60¢, $1, $1.40, $1.80, $2.20, $2.60, $3

G4. Can: 20¢, 40¢, 50¢, 60¢, 70¢, 80¢, 90¢ and $1. Cannot: 10¢ and 30¢. 20¢, 40¢, 60¢, 80¢ and $1 are made of 20¢ coins only. 50¢, 70¢ and 90¢ need one 50¢ coin and then 0, 20¢ or 40¢ more in 20¢ coins. 10¢ is smaller than every coin, and 30¢ is too small for a 50¢ coin and is not made of 20¢ coins.

G5. Open. Example: 50¢, 30¢, 20¢, 60¢, 40¢. The two baskets that cost $1 are 50¢ + 30¢ + 20¢ and 60¢ + 40¢. No other basket makes exactly $1, so the list has exactly two.

G6. Ben $3, Ali $5, Siti $7

G7. $20 ($5 + $5 = $10 before the pen; $10 + $10 = $20)

G8. Mei could be right (4 sweets and 4 lollipops: 20¢ + 80¢ = $1.00). Siti cannot. All the amounts: 55¢, 70¢, 85¢, $1.00, $1.15, $1.30, $1.45 (1 to 7 lollipops). They go up by 15¢ each time, because swapping one sweet for one lollipop adds 15¢. $1.50 is not on the list.

G9. 15 amounts: $1, $2, $3, $5, $6, $7, $8, $10, $11, $12, $13, $15, $16, $17, $18

Lesson 7: Fair Shares

S1. 2/5

S2. 4

S3. 1/3 (3 parts are bigger than 6 parts)

S4. 7/10

S5. 3/8

S6. 2/6

T1. 1/12, 1/5, 1/3, 1/2

T2. No. Eighths are smaller than fifths (the more equal parts, the smaller each part). So 3 eighths is less than 3 fifths: 3/5 is bigger than 3/8.

T3. No. 3 eighths and 2 eighths make 5 eighths: 5/8. The parts are still eighths, like 3 apples and 2 apples make 5 apples. Adding the bottom numbers would change the size of the parts.

T4. Yes, if his pizza is bigger. Halves of different-size wholes are different sizes.

T5. Any 3 of: 4 strips across, 4 strips down, a 2 by 2 grid, both lines from corner to corner (4 same triangles). Every part must be the same size.

T6. 1/8 more; 3/8 is left (5 eighths eaten).

T7. 8 parts; each part is one eighth (1/8). 2 eighths make 1 quarter.

T8. 3/12 (9 twelfths shaded, 12 – 9 = 3)

G1. 8 equal parts; each is 1/8 (1 fold: 2 parts, 2 folds: 4, 3 folds: 8).

G2. 1/4. After Siti, half is left; Ravi eats half of that half, which is a quarter of the pizza, so a quarter is left.

G3. 6/9 and 3/9 (take away the extra 3 ninths: 6 left; share: 3 and 3; give the 3 back: 6 and 3)

G4. 3 ways: 1/6 + 5/6, 2/6 + 4/6, 3/6 + 3/6

G5. Open. Example: Ali eats 4/10, Mei eats 3/10, Ben eats 1/10 and 2/10 is left: 4 + 3 + 1 + 2 = 10 tenths = 1 whole. Only two sets of three different numbers of pieces add up to 8: 1, 3 and 4, or 1, 2 and 5.

G6. No. 1/8 has a bigger bottom number than 1/4 but is smaller: the bar cut into 8 equal parts has shorter parts than the bar cut into 4 equal parts. When the bottom numbers are the same, the parts are the same size, so the fraction with more parts (the bigger top number) is bigger, e.g. 5/9 is bigger than 3/9.

G7. 3/12 (and Ali gets 5/12): 8 pieces are shared; take away Ali's extra 2, share 6 as 3 and 3.

G8. 4 ways (tenths): 1, 2, 7; 1, 3, 6; 1, 4, 5; 2, 3, 5.

G9. 1/5 and 1/6

Lesson 8: Shape Builders

S1. 4

S2. 2

S3. 6

S4. 5 (4 small + 1 big)

S5. 9

S6. A cylinder

T1. 19 (12 + 6 + 1)

T2. 9: four small squares (1 by 1), two long ones (1 high and 2 long), two tall ones (2 high and 1 long), and the whole grid (2 by 2). 4 + 2 + 2 + 1 = 9.

T3. E folds into a cube (it is a staircase of squares, one of the 11 cube nets). F does not: the row of 5 squares wraps round and the 1st and 5th squares land on the same face.

T4. No. When you fold the line round, the 5th square lands on the same face as the 1st and the 6th on the 2nd, and the two ends are left open.

T5. No straight sides and 4 curved sides.

T6. 6 (3 of length 1, 2 of length 2, 1 of length 3)

T7. 8 (18 − 6 = 12, then 12 − 4 = 8)

T8. 16 (8 in each layer)

G1. 20 (12 + 6 + 2). Also a 7 by 2 grid.

G2. 8 (4 small + 4 made of 2 small ones)

G3. Open. Examples: a stack 5 long, 2 wide and 2 high is 5 × 2 × 2 = 20; or four layers of 9, 6, 4 and 1 cubes (a 3 by 3, a 3 by 2, a 2 by 2 and a 1 by 1 layer: 9 + 6 + 4 + 1 = 20). Check that every layer sits on a layer at least as big.

G4. 1 (only the cube in the very middle is not on the outside)

G5. 18 (9 + 6 + 3)

G6. 20 (10 sit on the bottom line + 10 sit on the line across)

G7. 30 (1 + 4 + 9 + 16)

G8. 55 (25 + 16 + 9 + 4 + 1)

G9. Paint on 3 faces: 8 (the corners). On 2 faces: 12 (one in the middle of each edge). On 1 face: 6 (the middle of each face). No paint: 1 (the centre). 8 + 12 + 6 + 1 = 27. For a 4 by 4 by 4 cube the unpainted cubes make a 2 by 2 by 2 block inside: 8 cubes.

Lesson 9: Measure Masters

S1. 600 g (300 + 300)

S2. 3 ℓ (4 + 3 = 7, and 10 − 7 = 3)

S3. 65 cm

S4. 6:10 pm

S5. pm and pm

S6. 35 minutes (count in 5s: 5, 10, 15, 20, 25, 30, 35)

T1. 1 apple balances 2 lemons. Swap the pomelo for 5 lemons: 2 apples and 1 lemon balance 5 lemons. Take 1 lemon off both sides: 2 apples balance 4 lemons, so 1 apple balances 2 lemons. A pomelo and an apple balance 5 + 2 = 7 lemons.

T2. 1 hour 25 minutes (10:45 to 11:00 is 15 minutes, 11:00 to 12:00 is 1 hour, 12:00 to 12:10 is 10 minutes).

T3. Apple 100 g, pomelo 300 g (5 apples balance 500 g).

T4. Fill 7, pour into 2, empty 2, pour again: 3 ℓ is left in the 7 ℓ jug.

T5. 3:30 pm (count back 15 minutes to 4:00, then 30 more minutes to 3:30)

T6. Buses: 6:50, 7:05, 7:20, 7:35, 7:50. He waits 10 minutes for the 7:50 bus and reaches the station at 8:15 am (7:50 to 8:00 is 10 minutes, 15 more minutes).

T7. 86 cm (30 + 30 + 30 = 90; there are 2 joins, so take away 2 + 2 = 4)

T8. 10:20 am (20 ℓ more is 4 lots of 5 ℓ, which takes 4 lots of 10 minutes = 40 minutes)

T9. 200 g (the books are 1000 − 200 = 800 g; 200 + 200 + 200 + 200 = 800)

T10. 2:55 pm

G1. 300 g (the 2 extra apples are 200 g)

G2. 6 minutes (4 pieces need only 3 cuts)

G3. 10:40 am (1 hour before 12:20 pm is 11:20 am, then 40 minutes earlier is 10:40 am)

G4. 11:40 am (4 hours, 5 minutes slow each hour: 20 minutes slow)

G5. Open. Example: a 3 ℓ and a 7 ℓ jug make 5 ℓ (see the pours in the question above). With a 4 ℓ and a 6 ℓ jug you can only ever measure 2 ℓ, 4 ℓ or 6 ℓ: both jugs hold an even number of litres, so every pour, fill or empty changes the amounts by even numbers. 5 is odd, so it is impossible.

G6. One way: fill the 5 ℓ jug and pour it into the 3 ℓ jug until that is full: 2 ℓ is left. Pour the 2 ℓ into the bucket. Then fill the 5 ℓ jug and pour it all into the bucket: 2 + 5 = 7 ℓ. (Other ways work too, e.g. 1 ℓ + 3 ℓ + 3 ℓ. Whole jugs alone give 3, 5, 6, 8, 9, 10 ..., never 7.)

G7. (3 ℓ, 7 ℓ): fill 7 (0, 7); pour (3, 4); empty 3 (0, 4); pour (3, 1); empty 3 (0, 1); pour (1, 0); fill 7 (1, 7); pour until the 3 ℓ jug is full (3, 5). 5 ℓ in the 7 ℓ jug.

G8. Take one ball off both sides of scale 1: 2 cubes balance 1 ball. On scale 2, swap the ball for 2 cubes: 3 cubes balance 300 g, so a cube is 100 g and a ball is 200 g. Check scale 1: 100 + 100 + 200 = 400 = 200 + 200.

G9. 10:20 am (8:15 to 8:50, break to 9:00, 9:00 to 9:35, break to 9:45, 9:45 to 10:20)

Lesson 10: Challenge Day

S1. 951 and 159

S2. 8 children

S3. Mei, by 1 (Ali 15, Mei 16)

S4. 50¢ + 20¢ + 10¢ (the only way)

S5. 3/8

S6. Siti is the tallest. Mei is the shortest.

T1. 1 picture (30 – 10 – 15 = 5 children)

T2. Day 6. Mei's totals: $3, $9, $18, $30, $45, $63. Ali's totals: $10, $20, $30, $40, $50, $60. On day 6 Mei has $63 and Ali has $60. (On day 5 Mei still has less: $45 and $50.)

T3. 16 (14 – 6 = 8 is half, 8 + 8 = 16).

T4. 49 will be in the pattern and 50 will not. The pattern adds 4 each time, so every number in it is 1 more than a number in the 4 times table: 5 = 4 + 1, 9 = 8 + 1, 13 = 12 + 1. 49 = 48 + 1, but 50 is 2 more than 48.

T5. 3 cows and 5 chickens (12 + 10 = 22)

T6. $20 (the half she kept is $4 + $6 = $10)

T7. Ali (Siti has blue, Ali is not red, so Ali is green and Ben is red)

T8. 6 pieces and 5 cuts

G1. Exactly the sets with a 0 card: 0, a, b make a0b, ab0, b0a, ba0. Sets without 0 make 6.

G2. Ali car, Mei bus, Siti walks, Ben MRT. No clue can go (8, 4 or 2 answers).

G3. Mei 2/12, Ben 4/12, Ali 6/12 (2 + 4 + 6 = 12). Ali's share is 6 slices × 100 g = 600 g. How to start: draw the three bars, take away the extras (2 + 4 = 6), share the other 6 equally (2 each), then put the extras back.

G4. Day 5. Totals so far: day 1 20¢, day 2 60¢, day 3 $1.20, day 4 $2.00 (exactly $2, not more), day 5 $3.00.

G5. Open. Check that the puzzle has exactly one answer (try to find a second one), that the hint names a way to start, and that the answer is checked against every clue. Example: "Ali has 5 coins, all 10¢ or 20¢, worth 90¢. How many of each?" The answer is one 10¢ and four 20¢ (10 + 80 = 90), and it is the only one, because each swap of a 10¢ for a 20¢ adds exactly 10¢.

G6. 3/10 is left; 15 cm

G7. Swimming 20, football 30 (2 pictures stand for 10 children, so 1 picture stands for 5)

G8. Ali 10¢, Mei 20¢, Ben 50¢, Siti $1. Ben is not $1 and two people have less than him, so Ben has 50¢; then Ali 10¢ and Mei 20¢, and Siti has the $1.

G9. 4 plates of chicken rice and 3 bowls of noodles ($16 + $9 = $25). Table: 0 plates → $21, 1 → $22, 2 → $23, 3 → $24, 4 → $25 ✓. Smallest $21 (all noodles), biggest $28 (all chicken rice), so $30 is impossible. Each swap of a bowl for a plate adds exactly $1, so every whole dollar from $21 to $28 is possible.